

A scalar first-order necessity theorem for limiting Sobolev inequalities on stratified groups

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Status. Candidate partial result, likely valid, pending human review. The result settles the source question only for scalar first-order operators.

1 Source question

Van Schaftingen and Yung study a stratified homogeneous group G with first layer $\mathfrak{g}_1$, homogeneous dimension Q , and a homogeneous left-invariant operator $A(D)$ of order k . Their endpoint estimate is

$$\left\| D^{k-1}u \right\|_{L^{Q/(Q-1)}(G;V)} \leq C \|A(D)u\|_{L^1(G;E)}, \quad u \in C_c^\infty(G;V). \quad (1)$$

They prove it from (i) maximal hypoellipticity and (ii) the existence of a compatible homogeneous operator $L(D)$ whose symmetrization is cocanceling [1]. On PDF page 4 they ask whether validity of (1) implies these conditions on a general stratified homogeneous group.

The three source crops below give equation (1.7), conditions (i)–(ii), and the open-question paragraph, respectively.

vector X_i , $i = 1, \dots, m$, gives rise to a left-invariant vector field on G , which by abuse of notation we also denote by X_i . Let now k be a positive integer, and $\mathcal{I}_k$ be the index set $\{1, \dots, m\}^k$. For $\gamma = (\gamma_1, \dots, \gamma_k) \in \mathcal{I}_k$, we write X_γ for the left-invariant differential operator on G given by

$$X_\gamma := X_{\gamma_1} \dots X_{\gamma_k}. \quad (1.6)$$

Note that X_γ depends on the ordering of the indices within γ if the group G is not abelian. If $A^\gamma \in \text{End}(V; E)$ for all indices $\gamma \in \mathcal{I}_k$, then

$$A(D) := \sum_{\gamma \in \mathcal{I}_k} A^\gamma X_\gamma,$$

defines a homogeneous left-invariant linear partial differential operator $A(D): C^\infty(G; V) \rightarrow C^\infty(G; E)$ of order k on G from V to E with real coefficients. We are interested in when the inequality

$$\|D^{k-1}u\|_{L^{Q/(Q-1)}(G;V)} \leq C \|A(D)u\|_{L^1(G;E)}, \quad (1.7)$$

holds for such an operator $A(D)$, for all $u \in C_c^\infty(G; V)$, where $D^{k-1}u := (X_\gamma u)_{\gamma \in \mathcal{I}_{k-1}}$. The inequality (1.7) is the natural generalization of (1.2) to stratified homogeneous groups.

If $A(D) = D$, then (1.7) is an endpoint Sobolev inequality which is known to hold [18, 19]. In the particular case, where G is a Heisenberg group, Baldi, Franchi and Pansu have proved that (1.7) holds when $A(D)$ is a (first- or second-order) operator of the Rumin complex [1, 2]; their proof relies on the structure of the Rumin complex and on a

Figure 1: The endpoint estimate (1.7), source PDF page 2.

Bourgain–Brezis duality estimate on stratified homogeneous groups [14] generalizing the Euclidean results [34–36].

Our main result asserts that for a homogeneous left-invariant linear partial differential operator $A(D): C^\infty(G; V) \rightarrow C^\infty(G; E)$ of order k as above, if

- (i) $A(D)$ is maximally hypoelliptic, that is, if there exists some $C > 0$ such that

$$\|D^k u\|_{L^2(G; V)} \leq C \|A(D)u\|_{L^2(G; E)}$$

for all $u \in C_c^\infty(G; V)$, and

- (ii) there exists a finite dimensional inner product space F , a positive integer ℓ and some linear homogeneous left-invariant partial differential operator $L(D) = \sum_{\lambda \in \mathcal{I}_\ell} B^\lambda X_\lambda$ of order ℓ on G from E to F (so each $B^\lambda \in \text{End}(E; F)$) such that the symmetrized operator $\text{Sym}(L)(D)$ is cocanceling, that is

$$\bigcap_{\xi \in \mathbb{R}^m} \left\{ e \in E : \sum_{\lambda \in \mathcal{I}_\ell} \xi^\lambda B^\lambda[e] = 0 \right\} = \{0\},$$

then (1.7) holds for all $u \in C_c^\infty(G; V)$ (see Theorem 4.1 below). Under the above hypotheses (i) and (ii) on $A(D)$, we also obtain Hardy–Sobolev inequalities of the form

$$\left\| \frac{D^{k-\ell} u}{\|x\|^\ell} \right\|_{L^1(G; V)} \leq C \|A(D)u\|_{L^1(G; E)} \quad (1.8)$$

where $\|x\|$ is a homogeneous norm on G and $\ell \in \{1, \dots, \min\{k, Q-1\}\}$ (see Theorem 5.1(a)).

Figure 2: The sufficient conditions (i)–(ii), source PDF page 3.

$$\left\| \frac{D^{k-1} u}{\|x\|^\ell} \right\|_{L^p(G; V)} \leq C_p \|D^k u\|_{L^p(G; V)},$$

which together with (1.9) implies (1.11).) The estimates (1.7) and (1.8) are then respectively a limiting case of (1.10) and (1.11) as $p \rightarrow 1^+$. It may be worth pointing out that the maximal hypoellipticity condition on $A(D)$ is actually equivalent (under our other assumptions on $A(D)$) to a hypoellipticity condition on $A^t(D)A(D)$; see again Theorem 3.3. It is an interesting open question whether the validity of (1.7) for all $u \in C^\infty(G; V)$ implies the conditions (i) and (ii) on $A(D)$ on a general stratified homogeneous group G .

In order to apply our main theorems, given the operator $A(D)$ one must construct a compatible $L(D)$ as in condition (ii) above. In Proposition 6.1 we develop a robust way of doing so that works in many examples of interest. In particular, in Proposition 7.2 below, we obtain the Gagliardo–Nirenberg–Sobolev type estimate

$$\|D^{k-1} u\|_{L^{Q/(Q-1)}(G)} \leq C \sum_{i=1}^m \|X_i^k u\|_{L^1(G)}$$

Figure 3: The open question, source PDF page 4.

2 Definitions and scope

Write

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_r, \quad [\mathfrak{g}_i, \mathfrak{g}_j] \subseteq \mathfrak{g}_{i+j}, \quad Q = \sum_{j=1}^r j \dim \mathfrak{g}_j,$$

and let $X_1, \dots, X_m$ be a basis of $\mathfrak{g}_1$. We assume $Q \geq 2$. For a finite-dimensional inner-product space E and a linear map $T : \mathbb{R}^m \rightarrow E$, define the scalar first-order operator

$$A_T(D)u = T(Du) = T(X_1u, \dots, X_mu). \quad (2)$$

If $L(D)$ is a homogeneous left-invariant operator from E to another finite-dimensional space, it is *compatible* with $A_T(D)$ when $L(D)A_T(D) = 0$. Its symmetrization is *cocanceling* when

$$\bigcap_{\xi \in \mathbb{R}^m} \ker \text{Sym}(L)(\xi) = \{0\}.$$

3 Proof intuition

Suppose T loses a horizontal direction. Choose that direction as X_1 . Take a test function that remains compact in the x_1 coordinate, while all other exponential coordinates are expanded according to their homogeneous weights. The transverse volume is R^{Q-1} . The leading X_1 derivative is killed by T , and every derivative falling on the transverse cutoff costs R^{-1} . Consequently the right side of the endpoint estimate is only $O(R^{Q-2})$. The left side is of order $R^{(Q-1)/q} = R^{Q-2+1/Q}$ for $q = Q/(Q-1)$, giving a contradiction.

Thus T is injective. This already gives maximal hypoellipticity, since a finite-dimensional injective map is bounded below. For compatibility, use the source paper's explicit compatible cocanceling operator for the full horizontal gradient, transport it to $\text{ran } T$ through a left inverse, and add a pure-derivative block on a complement of $\text{ran } T$.

The argument does not address $k \geq 2$ or $\dim V > 1$.

4 The partial theorem

Theorem 1 (Scalar first-order characterization). *Let G be a stratified homogeneous group with homogeneous dimension $Q \geq 2$, let E be finite-dimensional, and let $A_T(D)$ be given by (2). Put $q = Q/(Q-1)$. The following are equivalent.*

1. *There exists $C < \infty$ such that*

$$\|u\|_{L^q(G)} \leq C \|A_T(D)u\|_{L^1(G;E)} \quad \text{for all } u \in C_c^\infty(G). \quad (3)$$

2. *The coefficient map $T : \mathbb{R}^m \rightarrow E$ is injective.*

3. *The operator $A_T(D)$ is maximally hypoelliptic and admits a compatible homogeneous $L(D)$ such that $\text{Sym}(L)(D)$ is cocanceling.*

In particular, the source question has an affirmative answer throughout the scalar first-order class.

Proof. We first prove (1) $\Rightarrow$ (2), which is the new necessity argument. Suppose instead that $0 \neq c \in \ker T$. Change the basis of $\mathfrak{g}_1$ so that $X_1 = c$. This only changes the equivalent norm used for Du . In the corresponding coordinates, $T(e_1) = 0$.

Complete $X_1, \dots, X_m$ to a homogeneous basis $Y_1 = X_1, Y_2 = X_2, \dots, Y_N$ of $\mathfrak{g}$, and let w_ν be the homogeneous weight of Y_ν . Thus $w_1 = \dots = w_m = 1$ and $\sum_{\nu=1}^N w_\nu = Q$. Use exponential coordinates of the first kind

$$(x_1, \dots, x_N) \mapsto \exp \left(\sum_{\nu=1}^N x_\nu Y_\nu \right).$$

Haar measure is a constant multiple of Lebesgue measure in these coordinates. The horizontal left-invariant fields have the standard polynomial form

$$X_j = \partial_{x_j} + \sum_{\nu: m < \nu \leq N} p_{j\nu}(x) \partial_{x_\nu}, \quad (4)$$

where $p_{j\nu}$ is a polynomial homogeneous of weighted degree $w_\nu - 1$.

Choose nonzero $\phi \in C_c^\infty(\mathbb{R})$ and nonzero $\eta \in C_c^\infty(\mathbb{R}^{N-1})$. For $R \geq 1$, set

$$\begin{aligned} \eta_R(x_2, \dots, x_N) &= \eta(R^{-w_2}x_2, \dots, R^{-w_N}x_N), \\ u_R(x) &= \phi(x_1)\eta_R(x_2, \dots, x_N). \end{aligned}$$

These are admissible compactly supported smooth functions. A change of variables gives

$$\|u_R\|_{L^q(G)} = c_{\phi, \eta} R^{(Q-1)/q} = c_{\phi, \eta} R^{Q-2+1/Q}. \quad (5)$$

We claim that, uniformly on $\text{supp } \phi \times \text{supp } \eta_R$,

$$|X_j \eta_R| \leq \frac{C}{R}, \quad j = 1, \dots, m. \quad (6)$$

The direct derivative $\partial_{x_j} \eta_R$ has this bound (and it is zero for $j = 1$). For a monomial x^α in $p_{j\nu}$, weighted homogeneity gives

$$\sum_{\mu=1}^N w_\mu \alpha_\mu = w_\nu - 1.$$

The coordinate x_1 stays in a fixed compact set, while $|x_\mu| = O(R^{w_\mu})$ for $\mu \geq 2$. Hence

$$|x^\alpha| = O \left(R^{\sum_{\mu \geq 2} w_\mu \alpha_\mu} \right) = O(R^{w_\nu - 1 - \alpha_1}).$$

Multiplication by $\partial_{x_\nu} \eta_R = O(R^{-w_\nu})$ gives $O(R^{-1 - \alpha_1})$, and therefore (6) follows from (4).

Since $X_j x_1 = \delta_{j1}$ and $T(e_1) = 0$,

$$\begin{aligned} A_T(D)u_R &= T(e_1)\phi'(x_1)\eta_R + \phi(x_1) \sum_{j=1}^m T(e_j)X_j \eta_R \\ &= \phi(x_1) \sum_{j=1}^m T(e_j)X_j \eta_R. \end{aligned}$$

The support has measure $O(R^{Q-1})$, so

$$\|A_T(D)u_R\|_{L^1(G;E)} \leq CR^{Q-2}. \quad (7)$$

Combining (5), (7), and (3) would give $R^{1/Q} \leq C'$ for every $R \geq 1$, a contradiction. Thus T is injective.

Next assume (2). Since T is an injective map between finite-dimensional spaces, there is $c > 0$ such that $|Tv| \geq c|v|$. Therefore

$$\|Du\|_{L^2(G)} \leq c^{-1} \|A_T(D)u\|_{L^2(G;E)}, \quad (8)$$

so $A_T(D)$ is maximally hypoelliptic. The endpoint inequality also follows directly from the horizontal Sobolev inequality (Proposition 7.1 of [1]):

$$\|u\|_{L^q(G)} \leq C_G \|Du\|_{L^1(G)} \leq C_G c^{-1} \|A_T(D)u\|_{L^1(G;E)}.$$

It remains to verify the compatibility part of (3). Proposition 7.1 of [1] constructs a homogeneous operator

$$L_{\nabla}(D) : C^\infty(G; \mathbb{R}^m) \longrightarrow C^\infty(G; F_0)$$

of some order s (their construction gives $s = r + 1$) such that

$$L_{\nabla}(D)D = 0, \quad \bigcap_{\xi \in \mathbb{R}^m} \ker \text{Sym}(L_{\nabla})(\xi) = \{0\}. \quad (9)$$

Let $E_0 = \text{ran } T$, choose a complement E_1 with $E = E_0 \oplus E_1$, let $P : E \rightarrow E_0$ be the associated projection, and let $S : E \rightarrow \mathbb{R}^m$ equal T^{-1} on E_0 and zero on E_1 . Define an order- s operator on E_1 by

$$H(D)h = (X_1^s h, \dots, X_m^s h).$$

Finally set

$$L(D)f = (L_{\nabla}(D)Sf, H(D)(I - P)f). \quad (10)$$

It is homogeneous of order s , and (9) gives

$$L(D)A_T(D)u = (L_{\nabla}(D)Du, 0) = 0.$$

If $e = e_0 + e_1 \in E_0 \oplus E_1$ lies in $\ker \text{Sym}(L)(\xi)$ for every ξ , the first block and (9) give $Se_0 = 0$, hence $e_0 = 0$. The second block has symbol

$$\text{Sym}(H)(\xi)e_1 = (\xi_1^s e_1, \dots, \xi_m^s e_1),$$

whose common kernel is zero, so $e_1 = 0$. Thus $\text{Sym}(L)$ is cocanceling and (2) implies (3). Finally, (3) implies (1) by the source paper's main endpoint theorem [1, Theorem 4.1]. This completes the equivalence. $\square$

5 Verification and limitations

The exponent gap is strict:

$$\frac{Q-1}{Q/(Q-1)} - (Q-2) = \frac{1}{Q} > 0.$$

The proof uses no numerical or symbolic computation. The main review point is the standard polynomial form (4) and its use under the partial dilation; the monomial calculation above makes the required uniform bound explicit.

The theorem assumes scalar domain $V = \mathbb{R}$, order $k = 1$, and $Q \geq 2$. The source question remains open here for higher order and for vector-valued domains. In those cases, a kernel of the representation symbol need not reduce to a single missing weight-one coordinate, so the present transverse cutoff does not settle the Rockland condition.

6 Novelty check

A bounded check was performed on 11 August 2026 using the run’s registry, solution, attempt, and proof-gap indexes; the locally loaded arXiv corpus; and the exact-title OpenAlex citation graph. The graph listed three citing works from 2023–2025. The locally available endpoint-Sobolev lecture notes discuss the Euclidean necessity theorem and cite [1] for stratified-group estimates, while the 2025 citing paper treats particular Laplacians on the Cartan group. No explicit general or scalar-first-order necessity result was found. This is not an exhaustive MathSciNet or Zentralblatt search, so the novelty verdict is moderate-confidence; the packet is submitted for human review as a candidate partial result.

References

- [1] J. Van Schaftingen and P.-L. Yung, *Limiting Sobolev and Hardy inequalities on stratified homogeneous groups*, Ann. Fenn. Math. 47 (2022), no. 2, 1065–1098; arXiv:2007.14532; DOI: 10.54330/afm.120959.